

A2. Demonstrate to create a project in remote repository and apply fork, merge, diff, merge conflict, branch and pull request concepts on repository using GitHub

Git **fork** command: git fork

Description

A **fork** creates a copy of someone else's GitHub repository into your account. It allows you to freely make changes without affecting the original project.

Syntax : git fork <https://github.com/HarshithK-froxcy/lab-1.1>

URL of Repository: <https://github.com/HarshithK-froxcy/lab-1.1>

Git **merge** command:

Description

The git merge command integrates changes from one branch into another. It combines histories, creating a merge commit if there are divergent changes. Use it to incorporate feature branches into the main branch after development.

Syntax : git merge <branch-name>

Example : git checkout main

git merge branch1

Git **diff** command:

Description

Shows the differences between two commits, branches, files, or your working directory and staging area.

Syntax : git diff (unstaged changes vs. index)

git diff --staged (staged vs. last commit)

git diff <commit1> <commit2> (between commits)

git diff <branch1> <branch2> (between branches)

Example : git diff

Git **branch** command:

Description

The git branch command manages branches: create, list, delete, or switch. Branches allow parallel development without affecting the main codebase.

Syntax : git branch <branch-name>

Example : git branch b1

git checkout b1

Git **merge conflict** demonstration :

Description

A merge conflict occurs when Git can't automatically resolve differences between branches (e.g., same file edited in conflicting ways). Git marks conflicts in files with

```

Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<<< HEAD (Current Change)
int c= a-b;
return a
=====
int c= a+b;
return d
>>>>>> 2995b777a69a4329a3566af2793c18dd1f4ec4b7 (Incoming Change)
  
```

Create **pull request** demonstration:

Description

A pull request (PR) on GitHub proposes changes from a branch (often in a fork) to the target repo's branch. It allows review, discussion, and merging. Use it for collaborative contributions.

Example:

The screenshot shows the GitHub interface for a repository named 'lab-1.1' by user 'HarshithK-froxy'. The repository is public and has 2 branches and 0 tags. A yellow notification bar at the top states 'assign had recent pushes 21 seconds ago' with a 'Compare & pull request' button. Below this, there's a section for 'Add files via upload' with a list of files: 'A1 (har).docx' (2 days ago), 'add.py' (2 days ago), and 'har.txt' (3 weeks ago). A 'README' file is also listed. On the right side, there are statistics for the repository: 0 stars, 0 forks, and 0 releases. The 'Releases' section is partially visible at the bottom right.